# Chapter 2. The Method of Mathematical Induction

*One Step at a Time*

## Contents


**2.1 Weak induction** **2**

**2.2 Strong induction** **3**

**2.3 Selected exercises** **3**

## 2.1 Weak induction

**Motivation.** Check the first case and show each case implies the next; the statement itself must be guessed.

**The method.** To prove $S(m)$ for all natural $m$:

1. *Initial step:* prove $S(m)$ for the smallest $m$ for which it is defined (usually 0 or 1).
2. *Induction step:* prove that $S(n)$ (the *induction hypothesis*) implies $S(n+1)$.

*Validity.* If $m+1$ is the smallest failure, it is not the first index, so $S(m)$ holds and the induction step gives $S(m+1)$.

**Example 2.1**

For all positive integers $m$,

$$1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6}. \tag{2.1}$$

*Solution.* $m = 1$: both sides are 1. Passing from $n$ to $n+1$ adds $(n+1)^2$ on the left and

$$\frac{(n+1)(n+2)(2n+3) - n(n+1)(2n+1)}{6} = \frac{(n+1)\big[(n+2)(2n+3) - n(2n+1)\big]}{6} = (n+1)^2$$

on the right.

**Example 2.2**

Let $f(m)$ be the maximum number of regions into which $m$ straight lines divide the plane. Then $f(m) = \frac{m(m+1)}{2} + 1$.
*Solution.* $f(1) = 2$. The last of $n+1$ lines crosses $k \le n$ others, so it splits $k+1$ regions: $f(n+1) \le f(n) + n + 1$, with equality in general position. Thus $f(n+1) = f(n) + n + 1 = \frac{n(n+1)}{2} + 1 + n + 1 = \frac{(n+1)(n+2)}{2} + 1$.

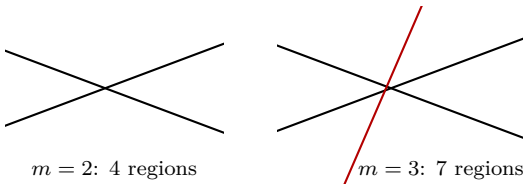



**Figure.** *Example 2.2: the red line splits 3 old regions, so $f(3) = f(2) + 3$.*

The formula is often guessed from small cases:

**Example 2.3**

Let $a_0 = 1$ and $a_{n+1} = 3a_n + 1$. Then $a_m = \frac{3^{m+1}-1}{2}$.
*Solution.* Guessed from $1, 4, 13, 40, 121$. True for $m = 0$, and $a_{n+1} = 3 \cdot \frac{3^{n+1}-1}{2} + 1 = \frac{3^{n+2}-1}{2}$.

A formula like $a_{n+1} = 3a_n + 1$ is a **recurrence relation**; $a_n = \frac{3^{n+1}-1}{2}$ is **explicit** and **closed** (no $\sum$ or $\prod$ over a varying number of terms); $\sum_{i=1}^{n} i$ is explicit but not closed, $\frac{n(n+1)}{2}$ is closed.

**Sets.** A finite **set** is an unordered collection of distinct objects ($\{1,3,2\} = \{2,1,3\}$); with repetitions, a **multiset**. $B \subseteq A$: every element of $B$ lies in $A$. The canonical $n$-element set is $[n] := \{1,2,\dots,n\}$.

**Theorem 2.4**

For every positive integer $n$, $[n]$ has $2^n$ subsets.

**Proof.** $[1]$ has $\emptyset, \{1\}$. Subsets of $[n+1]$ without $n+1$ are the $2^n$ subsets of $[n]$; those with $n+1$ are $\{n+1\} \cup T$, $T \subseteq [n]$: another $2^n$. ■

**Pitfalls**

"$2n+1$ is even" has a valid induction step but no initial case. "All horses have the same colour" fails because the step from 1 to 2 (two disjoint groups of one horse) is invalid: the step must hold for *every* $n$.

## 2.2 Strong induction

**Motivation.** When $a_{n+1}$ depends on all earlier terms, the induction step may use *all* smaller values.

**Example 2.5**

Let $a_0 = 0$ and $a_{n+1} = a_0 + a_1 + \cdots + a_n + n + 1$ for $n \geq 0$. Then $a_n = 2^n - 1$.
*Solution.* True for $n = 0$. If $a_k = 2^k - 1$ for all $k \leq n$, then

$$a_{n+1} = \sum_{k=0}^{n} (2^k - 1) + n + 1 = (2^{n+1} - 1) - (n+1) + n + 1 = 2^{n+1} - 1.$$

**The method (strong induction).**

1. *Initial step:* prove the statement for the smallest $n$ for which it is defined.
2. *Induction step:* prove that the statement for all integers $< n+1$ implies it for $n+1$.

*Validity.* As for weak induction, via the smallest failure. From now on $\mathbb{N} = \{0,1,2,\dots\}$.

**Example 2.6**

Let $f(0) = 1$, $f(1) = 2$ and $f(n+1) = f(n-1) + 2f(n)$ for $n \geq 1$. Then $f(n) \leq 3^n$ for all $n \in \mathbb{N}$.
*Solution.* True for $n = 0, 1$; then $f(n+1) \leq 3^{n-1} + 2 \cdot 3^n = 7 \cdot 3^{n-1} < 3^{n+1}$ (using both $n-1$ and $n$).

**Variations**

One may shift the parameter ($n - 1 \Rightarrow n$) or induct over even or odd integers only.

## 2.3 Selected exercises

**Motivation.** Book exercises: the work lies in choosing the statement and what to remove.

**Exercise 2.1 (Iterated differences of a polynomial)**

For a polynomial $p$ let $\Delta p(n) = p(n) - p(n-1)$, and let $\Delta^k p$ be the result of applying $\Delta$ $k$ times ($\Delta^0 p = p$). Prove that if $p$ has degree $d$, then $\Delta^d p$ is a constant.

**Solution.** Claim: if $\deg p = d$ with leading coefficient $a \neq 0$, then $\Delta^d p = d!\, a$.

*Step 1 (one difference lowers the degree).* For $j \geq 1$, by Theorem 4.1

$$n^j - (n-1)^j = n^j - \sum_{i=0}^{j} \binom{j}{i} n^i (-1)^{j-i} = j\, n^{j-1} + (\text{terms of degree} \leq j-2),$$

and for $j = 0$ it is 0. With $p(n) = \sum_{j=0}^{d} c_j n^j$, $c_d = a$,

$$\Delta p(n) = \sum_{j=1}^{d} c_j (n^j - (n-1)^j) = d\, a\, n^{d-1} + (\text{terms of degree} \leq d-2).$$

*Step 2 (induction on $d$).* $d = 0$: $\Delta^0 p = a = 0!\, a$. For $d \geq 1$, $q = \Delta p$ has degree $d-1$ and leading coefficient $da$, so $\Delta^d p = \Delta^{d-1} q = (d-1)! \cdot da = d!\, a$. ■

**Exercise 2.2 (A player who lists everybody)**

At a tennis tournament every two players played each other exactly once. Afterwards each player listed the names of those he defeated and of those defeated by someone he defeated. Prove that some player listed the names of everybody else.

**Solution.** *$x$ covers $y$* if $x$ beat $y$ or $x$ beat some $z$ who beat $y$. Induction on the number $n$ of players.

*Step 1 (initial step).* $n = 1$: vacuous.

*Step 2 (induction step).* Fix $X$ among $n+1$ players; by hypothesis some $K$ covers the other $n$. If $K$ covers $X$, done.

*Step 3 (the remaining case).* Otherwise $X$ beat $K$ and every $z$ beaten by $K$ (else $K$ covers $X$). For $y \neq X$: if $y = K$ or $K$ beat $y$, then $X$ beat $y$; if $K$ beat $z$ who beat $y$, then $X$ beat $z$. So $X$ covers everybody. ■

**Exercise 2.15 (Arithmetic and geometric means)**

Prove by induction that for all nonnegative reals $a_1, \ldots, a_n$,

$$\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

**Solution.** Equivalently $a_1 \cdots a_n \leq A^n$, $A$ the mean; induct on $n$ ($n = 1$: equality).

*Step 1 (setting up).* For $a_1, \ldots, a_{n+1} \geq 0$ with mean $A$, reorder so $a_n = \min$, $a_{n+1} = \max$; then $a_n \leq A \leq a_{n+1}$.

*Step 2 (replacing two numbers).* $b = a_n + a_{n+1} - A \geq a_n \geq 0$ and

$$A\, b - a_n a_{n+1} = (A - a_n)(a_{n+1} - A) \geq 0, \qquad \text{so} \qquad a_n a_{n+1} \leq A\, b.$$

*Step 3 (applying the hypothesis).* $a_1, \ldots, a_{n-1}, b$ have sum $nA$, so $a_1 \cdots a_{n-1}\, b \leq A^n$ and

$$a_1 \cdots a_{n-1} a_n a_{n+1} \leq a_1 \cdots a_{n-1} \cdot A\, b \leq A^n \cdot A = A^{n+1}.$$

■

**Exercise 2.17 (An L-shape of 1's)**

(a) Let $m, n \geq 2$ and let $A$ be an $m \times n$ matrix with entries 0 and 1, at least $m+n-1$ of which are 1. Prove that some entry 1 has another 1 in its row and another 1 in its column.
(b) Does the statement remain true with only $m+n-2$ entries 1?

**Solution.**

(a) Otherwise every 1 is alone in its row ($R$) or, if not, alone in its column ($C$).

*Step 1 (counting).* $|R| \leq m$, $|C| \leq n$, $|R| + |C| \geq m+n-1$, so $|C| = n$ or $|R| = m$.

*Step 2 (case $|C| = n$).* Each column's only 1 lies in $C$, so $R = \emptyset$ and $n \geq m+n-1$, i.e. $m \leq 1$.

*Step 3 (case $|R| = m$).* Symmetrically $C = \emptyset$ and $n \leq 1$.

(b) No: 1's at $(1,1), \ldots, (1,n-1)$ and $(2,n), \ldots, (m,n)$ ($m+n-2$ ones); each first-row 1 is alone in its column, each last-column 1 alone in its row. ■